

Q8: Probability of Being a Girl?

Problem Statement

A family has two children. Assume each child is equally likely to be a boy or a girl, independently of the other.

What is the probability that both children are girls, given that **at least one** of them is a girl?

Solution

Step 1: Define the sample space

Let B denote a boy and G denote a girl. For two children, listing the older child first, the sample space consists of four equally likely outcomes:

$$\Omega = \{BB, BG, GB, GG\}.$$

Each outcome has probability $\frac{1}{4}$ since children are independent and equally likely to be boy or girl.

Step 2: Define the events

Let A be the event that at least one child is a girl:

$$A = \{BG, GB, GG\}.$$

Let B be the event that both children are girls:

$$B = \{GG\}.$$

Step 3: Compute the conditional probability

We want $P(B \mid A)$, the probability that both are girls given that at least one is a girl.

By definition of conditional probability:

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)}.$$

Since $B \subseteq A$, we have $B \cap A = B$. Therefore:

$$P(B \cap A) = P(B) = \frac{1}{4}.$$

The probability of A is:

$$P(A) = \frac{3}{4}.$$

Thus:

$$P(B \mid A) = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}.$$

Step 4: Interpretation

The answer is $\frac{1}{3}$, not $\frac{1}{2}$.

This is a classic example of conditional probability that often surprises people. The key insight is that knowing "at least one is a girl" eliminates only the BB outcome, leaving three equally likely possibilities: BG , GB , and GG . Only one of these three has both children being girls.

Common Mistake

Some might incorrectly think the answer is $\frac{1}{2}$ by reasoning: "If one child is a girl, the other is equally likely to be a boy or a girl." This would be correct if we knew which specific child (e.g., the older one) was a girl. However, the condition "at least one is a girl" does not specify which child, so the sample space is reduced differently.

Comparison

- $P(\text{both girls} \mid \text{older child is a girl}) = \frac{1}{2}$
- $P(\text{both girls} \mid \text{at least one is a girl}) = \frac{1}{3}$

Final Answer

$$\boxed{\frac{1}{3}}$$